Standard Normal CDF

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7703	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Example 5.6

Suppose that sample of size n=25 are selected at random from normal population with mean 100 and standard deviation 10. What is the probability that the sample mean falls in the interval from $\mu_{\bar{X}} - 1.7\sigma_{\bar{X}}$ to $\mu_{\bar{X}} + 1.5\sigma_{\bar{X}}$

$$z = \frac{\bar{x} - \mu_{\bar{X}}}{\sigma_{\bar{X}}}$$

$$n = 25, N \sim (100, 10^2), \mu_{\bar{x}} = 100, \sigma_{\bar{X}} = \sigma_X / \sqrt{n} = 10 / \sqrt{25} = 2$$

$$P\left[100 - 1.7(2) \le \bar{X} \le 100 + 1.5(2)\right] = P\left[96.6 \le \bar{X} \le 103\right]$$

$$= P\left[\frac{96.6 - 100}{2} \le Z \le \frac{103 - 100}{2}\right] = P[Z \le 1.5] - P[Z \le -1.7] = 0.8886$$

Example 5.7

$$\sigma_X^2 = \frac{(b-a+1)^2-1}{12} = \text{Variance of } X$$

Suppose that X has a discrete uniform distribution

$$P_X(x) = \begin{cases} 1/4 & x = 1,2,3,4 \\ 0 & otherwise \end{cases}$$

A random sample of n=36 is selected from this population. Find the probability that the sample mean is greater than 2.3 but less than 2.7.

$$n = 36, \mu_x = E[X] = 5/2, \sigma_X = \sqrt{\frac{5}{4}}, \sigma_{\bar{X}} = \sigma_X / \sqrt{n} = \frac{\sqrt{5/4}}{6}$$

$$P\left[2.3 < \bar{X} < 2.7)\right] = P\left[-1.0733 < Z < 1.0733\right] \qquad \sigma_{\bar{X}} = \sigma_X / \sqrt{n}$$

$$= P[Z < 1.0733] - P[Z < -1.0733] = 0.7169$$

Example 5.8

A bank teller serves customers standing in the queue one by one. Suppose that the service time X_i for customer i has mean $EX_i = 2$ (minutes) and $\mathrm{Var}(X_i) = 1$. We assume that service times for different bank customers are independent. Let Y be the total time the bank teller spends serving 50 customers. Find $P(90 < Y < 110)$.

$E(Y) = E(X_1 + X_2 + \ldots + X_{50}) = 50 \times 2 = 100$

$Var(Y) = Var(X_1 + X_2 + \ldots + X_{50}) = Var(X_1) + Var(X_2) \ldots + Var(X_2)$
$= 50 \cdot 1 = 50$

std of $Y \Rightarrow \sqrt{50}$

$$Y = X_1 + X_2 + \ldots + X_n,$$

where $n = 50$, $EX_i = \mu = 2$, and $\mathrm{Var}(X_i) = \sigma^2 = 1$. Thus, we can write

$$\begin{aligned} P(90 < Y \leq 110) &= P\left(\frac{90 - n\mu}{\sqrt{n}\sigma} < \frac{Y - n\mu}{\sqrt{n}\sigma} < \frac{110 - n\mu}{\sqrt{n}\sigma}\right) \\ &= P\left(\frac{90 - 100}{\sqrt{50}} < \frac{Y - n\mu}{\sqrt{n}\sigma} < \frac{110 - 100}{\sqrt{50}}\right) \\ &= P\left(-\sqrt{2} < \frac{Y - n\mu}{\sqrt{n}\sigma} < \sqrt{2}\right). \end{aligned}$$

By the CLT, $\frac{Y - n\mu}{\sqrt{n}\sigma}$ is approximately standard normal, so we can write

$$\begin{aligned} P(90 < Y \leq 110) &\approx \Phi(\sqrt{2}) - \Phi(-\sqrt{2}) \\ &= 0.8427 \end{aligned}$$

Estimation

- We begin by formally outlining the statistical inferences.
- Statistical inferences consists of methods making inferences or generalization about a population.
  - In this part, we shall use statistical inferences to estimate unknown population parameters such as the mean, the proportion and variance by computing statistics from random samples

Terminology

Def: **Estimation** is a kind of procedure to estimate parameters.

Def: **Estimator** is a function of random variable, that must be plausible (reasonable) for population parameter

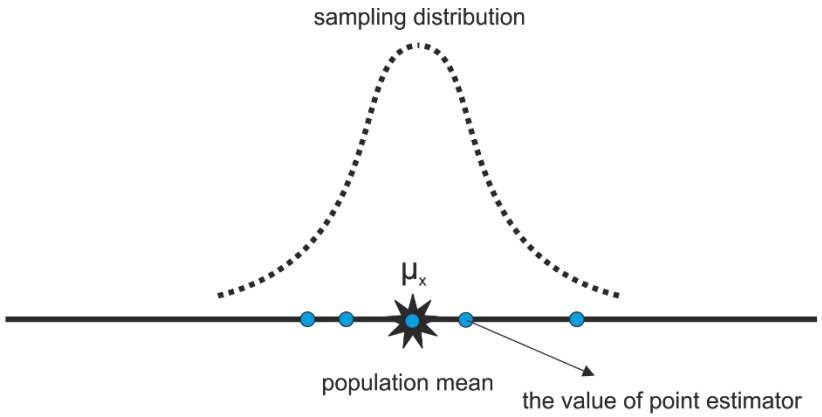
Def: **Estimand** is a paramater that is estimated by by estimator.

Estimator	Parameter to be estimated (estimand)
$f(x) \rightarrow$	μ_X
	σ_X
	σ_X^2

An estimate of a population parameter may be expressed in two ways

1. Point estimator: A point estimate of a population parameter is a single value of a statistic. A guess for the value of parameter. **The problem with using a single point (or value) is that it will very probably be wrong. In fact, with a continuous random variable, the probability that the variable is equal to a particular value is zero. So, $P(\bar{X}=\mu) = 0$.**

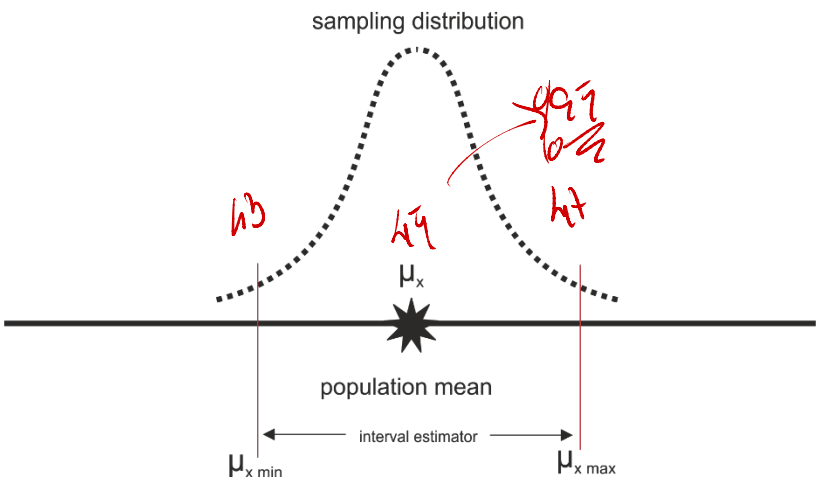
It is very unlikely that the parameter desired to be estimated will be the same to the value obtained from point estimators.



2. Interval estimator: An interval estimate is defined by two numbers.

It will give us an interval instead point value, by this means you are able to make a prediction with high accuracy than doing with point estimator.

For example, $\mu_{Xmin} < \mu_X < \mu_{Xmax}$ is an interval estimate of the population mean μ. It indicates that the population mean is greater than μ_{Xmin} but less than μ_{Xmax} with probability 0.95



Instructional Example: **Average height**
Point Estimation

Point estimate. A point estimate of a population parameter is a single value of a statistic.

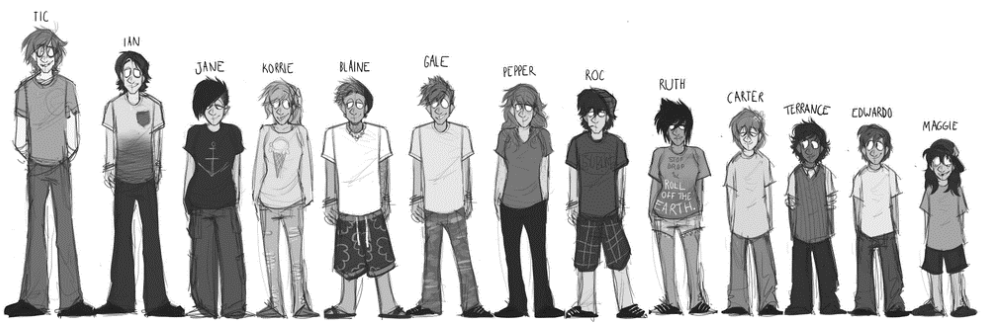
Example-1 Estimate the average height of students

- Suppose we are interested in the **average height** of students at Özyeğin University.
- The population size is very big. (approx. 8000 students)
- Take a random sample of 100 among all students.

Which functions would be plausible for average height?

$$f(x) = ????$$

What are the desirable property of "good" decision function that would influence us to choose one estimator rather than other?



Instructional Example

Sample the people and choose an estimator to get the best approximated result

- **Method 1**: The sample mean is the estimate of our estimand provided by the "sample average" estimator.

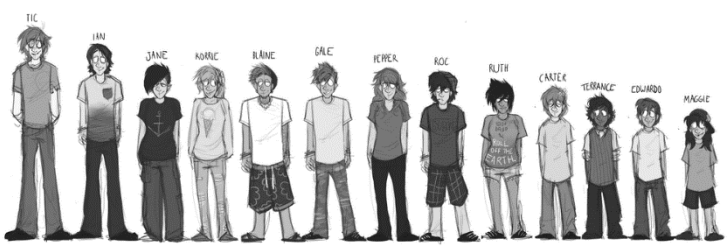
$$f(x)=\frac{\sum_{i=1}^{n} x_i}{n}$$

- **Method 2**: The sample median of the heights as an estimator.

$$f(x)= single\ middle\ value$$

- **Method 3**: The average of the tallest and shortest from the sample

$$f(x)=\frac{x_{min}+x_{max}}{2}$$



Which estimator is the best estimator?

Which estimator can predict more accurately?

Bias of a point estimator

A statistic $\hat{\Theta}$ is said to be an **unbiased estimator** of the parameter θ if

$$\mu_{\hat{\Theta}} = E(\hat{\Theta}) = \theta.$$

The bias is a kind of metric that evaluate the performance of the estimator.

The bias of an estimator is the expected value of the difference between the estimate and the true value

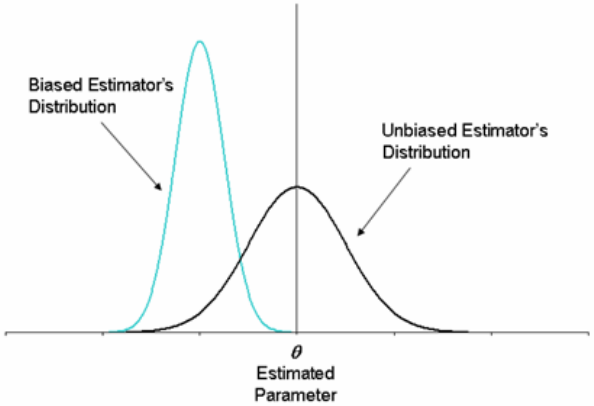
$$bias = E[estimator\ function - true\ value] = E[estimator\ function] - true\ value$$

- A good property of a point estimator is being unbiased

Types of Estimator

- Unbiased/biased estimator
  - Let $\hat{\theta}$ be point estimator function

    and θ the true value,
    - If $E[\hat{\theta}]-\theta = 0$, the estimator is unbiased estimator
    - If $E[\hat{\theta}]-\theta \neq 0$, the estimator is biased estimator



Bias of an estimator

The best way to understand whether an estimator is biased is to equalize the sample size to the population size. If the estimator is unbiased, then the sample mean is expected to be equal to population mean

- Method 1: Sample Mean (Unbiased)

$$f(x)=\frac{\sum_{i=1}^{n} x_i}{n} \xrightarrow{sample\ size=population\ size} \frac{\sum_{i=1}^{N} x_i}{N} \xrightarrow{\text{yields}} bias = E\left[\frac{\sum_{i=1}^{N} x_i}{N}\right] - \frac{\sum_{i=1}^{N} x_i}{N} = 0$$

- Method 2: Sample Median (Biased)

$$f(x)=\text{middle value} = \mathrm{C} \xrightarrow{sample\ size=population\ size} C \xrightarrow{\text{yields}} bias = E[C] - \frac{\sum_{i=1}^{N} x_i}{N} = C - \frac{\sum_{i=1}^{N} x_i}{N} \neq 0$$

- Method 3: (Biased)

$$f(x)=\frac{x_{min}+x_{max}}{2} = \mathrm{C} \xrightarrow{sample\ size=population\ size} C = \frac{x_{min}+x_{max}}{2} \xrightarrow{\text{yields}} bias = E[C] - \frac{\sum_{i=1}^{N} x_i}{N} = C - \frac{\sum_{i=1}^{N} x_i}{N} \neq 0$$

Example 5.9

Let X_1 and X_2 be two independent random variables with mean μ and variance σ^2 Suppose that we have two estimators of μ

$$\hat{\theta}_1 = \frac{X_1}{2} + X_2, \hat{\theta}_2 = \frac{X_1+3X_2}{4}$$

- Are both estimator unbiased estimator of μ? (No, the second estimator is unbiased)

$E\left(\frac{X_1}{2}\right)+E(X_2) = \mu/2 + \mu = 1.5\mu$

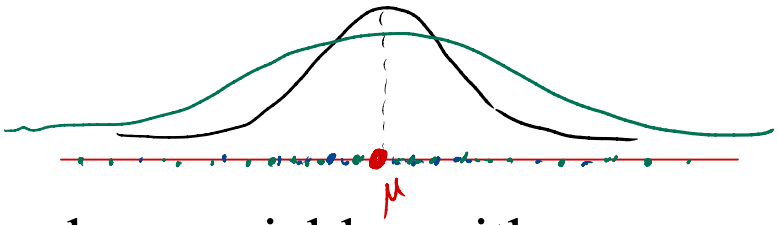
real mean

$$\mathrm{E}[\hat{\theta}_1] = \mathrm{E}\left[\frac{X_1}{2} + X_2\right] = 1.5\mu \xrightarrow{\textit{calculating bias}} 1.5\mu - \mu = 0.5\mu$$

$$\mathrm{E}[\hat{\theta}_2] = \mathrm{E}\left[\frac{X_1+3X_2}{4}\right] = \mu \xrightarrow{\textit{calculating bias}} \mu - \mu = 0$$

$E\left(\frac{X_1}{4}\right)+E\left(\frac{3X_2}{4}\right) = \mu/4 + 3\mu/4 = \mu$

Example 5.10



Let X_1 and X_2 be independent random variables with mean μ and variance σ^2 Suppose that we have two estimators of μ

$$\hat{\theta}_1 = \frac{X_1+X_2}{2}, \hat{\theta}_2 = \frac{X_1+3X_2}{4}$$

- Are both estimator unbiased estimator of μ? (Yes)

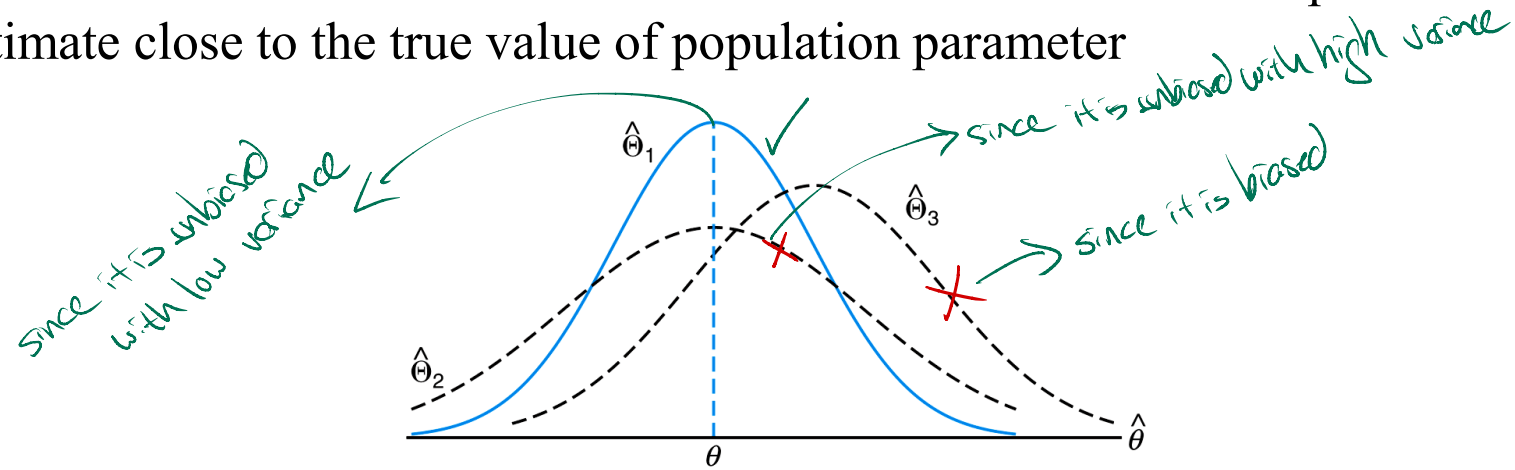
$$E[\hat{\Theta}_1] = E[\frac{X_1+X_2}{2}] = \frac{2\mu}{2} = \mu \xrightarrow{\text{calculating bias}} \mu - \mu = 0$$

$$E[\hat{\Theta}_2] = E[\frac{X_1+3X_2}{4}] = \frac{4\mu}{4} = \mu \xrightarrow{\text{calculating bias}} \mu - \mu = 0$$

Variance of an estimator

If we consider all possible unbiased estimators of some parameter θ, the one with the smallest variance is called the **most efficient estimator** of θ.

- Suppose that $\hat{\theta}_1$ and $\hat{\theta}_2$ are unbiased estimators.
- An estimator with **low variance is preferable over an estimator with high variance**
- The variance of these estimators show that which estimator produce an estimate close to the true value of population parameter



Example 5.10 (cont'd)

Let X_1 and X_2 be independent random variables with mean μ and variance σ^2 Suppose that we have two estimators of μ

$$\hat{\theta}_1 = \frac{X_1+X_2}{2}, \hat{\theta}_2 = \frac{X_1+3X_2}{4}$$

- What is the variance of each estimator?

$$Var\left(\frac{X_1+X_2}{2}\right) = Var\left(X_1/2\right) + Var\left(X_2/2\right)$$
$$= 1/4\, Var(X_1) + 1/4\, Var(X_2)$$
$$= \sigma^2/2$$

$$Var[\hat{\Theta}_1] = Var[\frac{X_1+X_2}{2}] = \frac{\sigma^2}{2}$$

$$Var[\hat{\Theta}_2] = Var[\frac{X_1+3X_2}{4}] = \frac{5\sigma^2}{8}$$

Which estimator is better? (The first estimator is better)